

MYCELIUM HOUSE

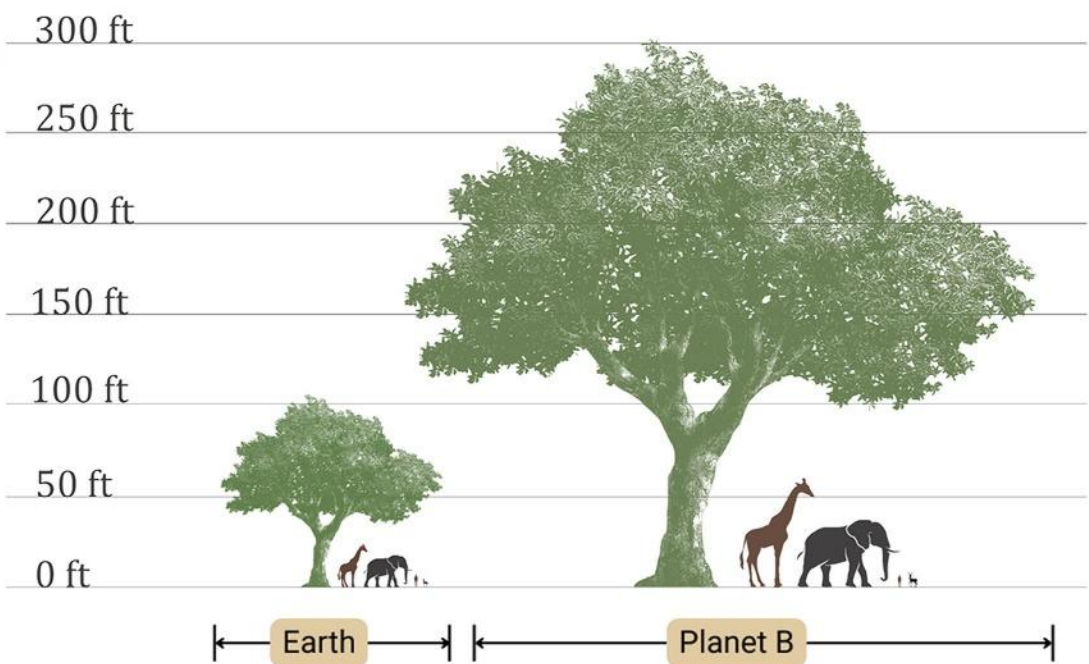


INTRODUCTION: Planet B: Humanity's Second Dawn

The Year 2300. Earth has reached the brink of collapse, no longer capable of sustaining human life. In humanity's final attempt to survive, Project Second Dawn is launched to search for a new home beyond the stars.

Among the discovered worlds, one planet stands apart—Planet B. Nearly identical to Earth in its atmosphere, flora, fauna, and ecosystems, it possesses one extraordinary difference: everything exists at three times Earth's scale.

To understand whether humanity can truly begin again, selected scientists and their families are sent to Planet B to study its environment, document its ecosystems, evaluate its habitability, and determine whether a family can not only survive, but build a home, a community, and a future. One such expedition is entrusted to Mr Vance and his family.



SITE : Tree branch



The site is a tree branch which is located within a dense forest on Planet B. Since all wildlife is three times larger than on Earth, the giant animals pose a constant threat to the scientists. For safety while preserving the natural ecosystem, the settlement is established within three massive trees. Each tree functions as a self-sufficient community, residential spaces, a community gathering area, a laboratory / resource and communication center.

Details



Site and Terrain: Tree branch in an invented alien forest biome, hypothetical site. Terrain is dense old-growth forest

Temperature: Temperate, mild year-round - no extreme seasonal swings

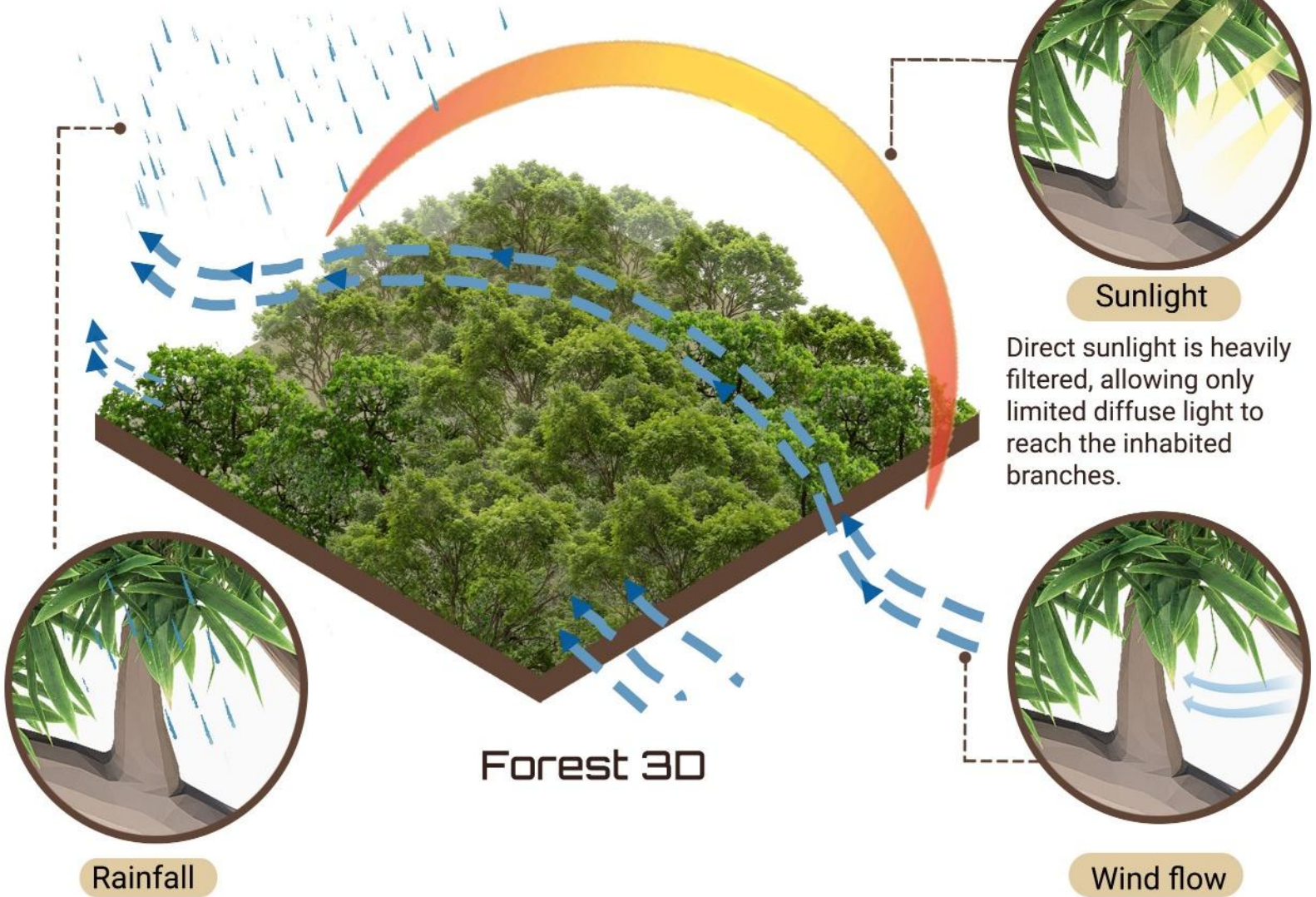
Rainfall: Moderate, evenly distributed through the year

Vegetation: Extremely tall host trees (~300 ft), thick bark, dense evergreen-like canopy cover

Fauna scale: Flora and fauna roughly 3× larger than Earth equivalents

Settlement spacing: Cluster of a few host trees, loosely spaced within the canopy, connected by bridges and short-range flight with backpack sized jets.

SITE ANALYSIS

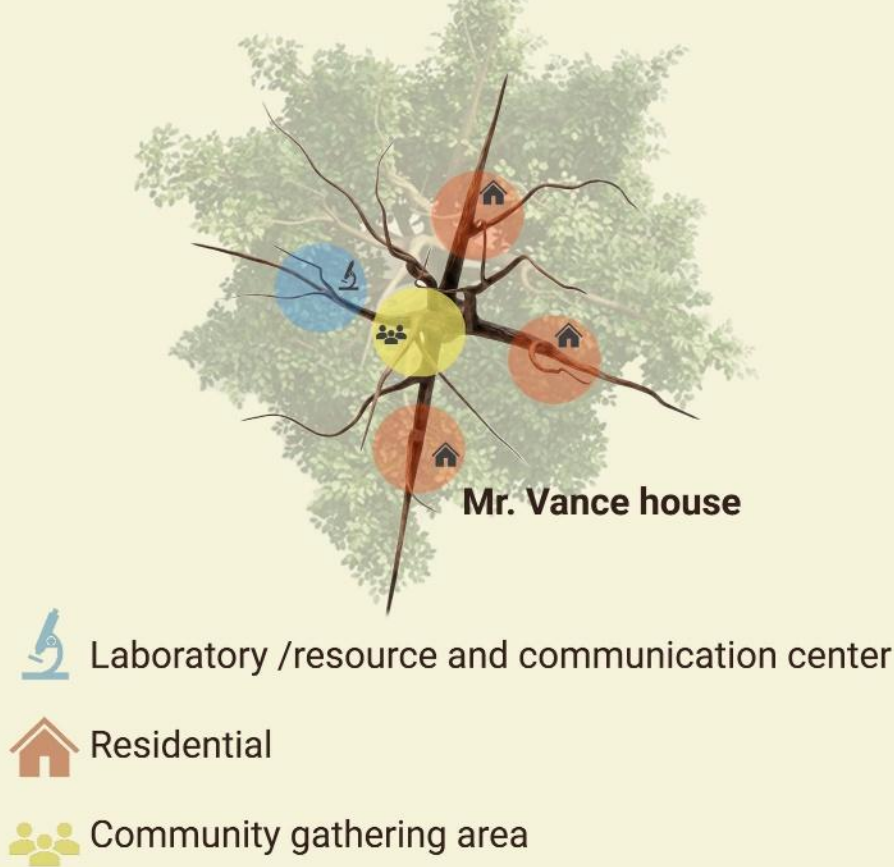


Rainfall
Rainfall is intercepted by the massive canopy before reaching the site. Water is released gradually through leaves and branches, with occasional concentrated drips from the large foliage.

The tree branch provides an ideal site, naturally shielding the habitat from heavy rain, strong winds, and giant wildlife while receiving filtered sunlight

Wind flow
The surrounding vegetation reduces wind speeds, as most airflow passes above the forest canopy. The branches naturally shield the habitat from strong winds, allowing greater facade permeability without prioritizing high wind resistance.

COMMUNITY



SWOT

- S** Flora and fauna are 3× larger — ground-level building is impractical, while elevated dwelling gives natural safety from large ground predators.
- W** Dense canopy disperses sunlight, limiting daylight at lower levels; also limits ground-level visibility and navigation.
- O** Elevation offers protection from ground threats and difficult terrain; Tree Also helps in camouflaging
- T** Animals being 3x larger makes any ground encounter dangerous, alongside fall risk from living at heights.

TAKEAWAYS

- Biomimetic design: The building should mimic natural forms to improve camouflage, reducing the likelihood of detection and attack by the giant wildlife.
- Maximize daylight: Use materials and facade systems that admit abundant filtered daylight into the interior.
- Reduced wind loads: As the forest canopy acts as a natural windbreak, structural design can prioritize openness and permeability over high wind resistance.
- Rainwater flow: Shape the building envelope to shed and redirect canopy runoff

MYCELIUM HOUSE

CONCEPT



Inspired by the ecological behaviour of mushrooms, the proposal explores a new relationship between architecture and nature. On Planet B, where every organism exists at three times Earth's scale, the house abandons the idea of dominating the landscape. Instead, it behaves as a quiet participant within the ecosystem, using biomaterial innovation, filtered light, and integrated spatial experiences to create a home that coexists with its environment. The project is not merely inspired by the form of a mushroom, but by its role within the forest: living with nature rather than competing against it.

CLIENT PROFILE

Mrs. Vance (40) – Microbiologist

Likes: Cultivating and examining small-scale organisms, tending plant specimens, hosting fellow researchers

Spaces needed: Garden (for specimens) and drawing room for hosting other scientists).

Mr. Vance (42) – Field Biologist

Likes: Fieldwork, cataloguing wildlife, quiet focused study, early mornings observing forest activity

Spaces needed: Quite study area for research work

Daughter (10) enjoys nature and art

Likes: Sketching plants and garden specimens, quiet time.

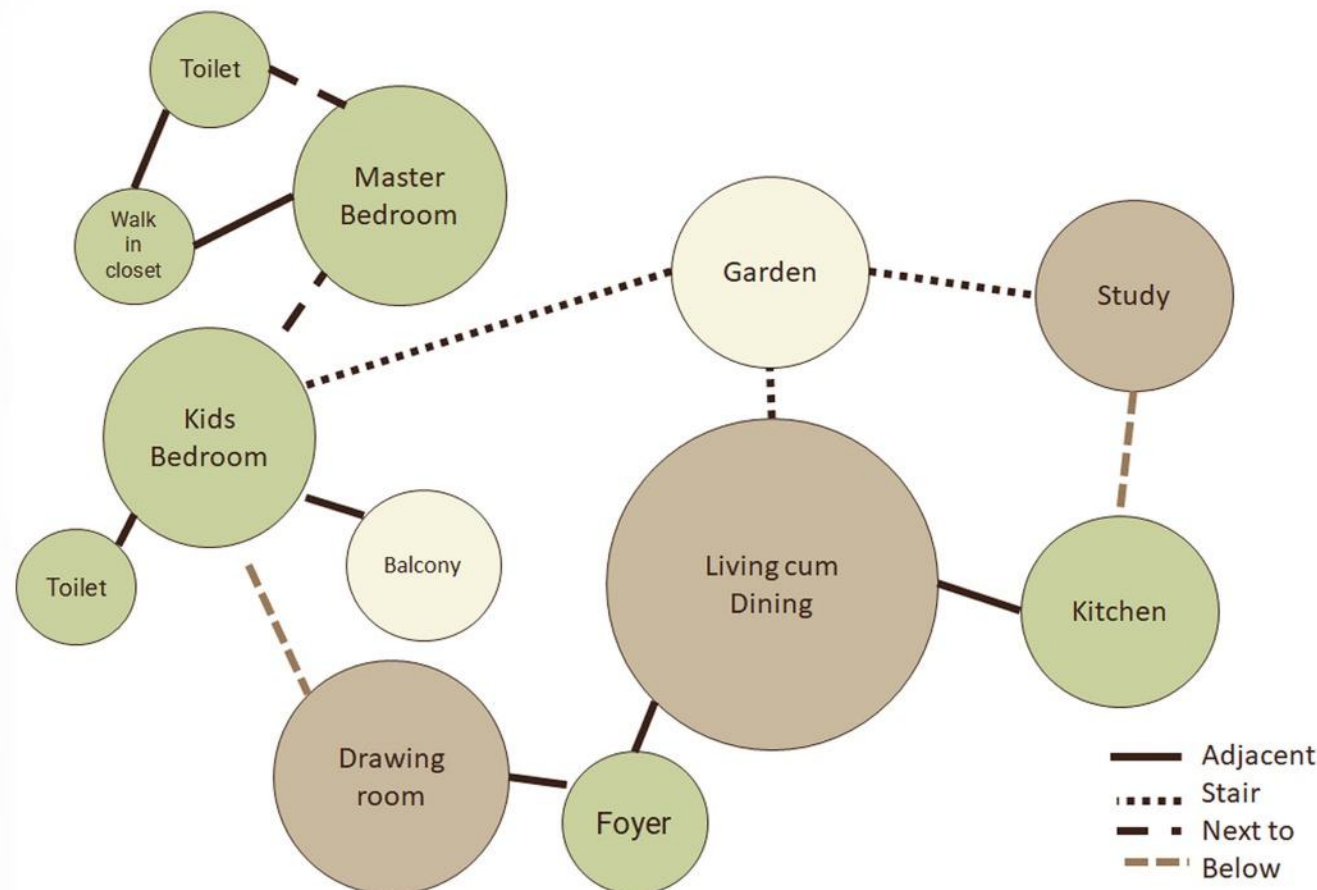
Spaces needed: Outdoor /indoor garden by the stair

Son (12) wildlife photography

Likes: Nature Photographing and observation

Spaces needed: View deck like balcony

BUBBLE DIAGRAM



FORM DEVELOPMENT

Problems

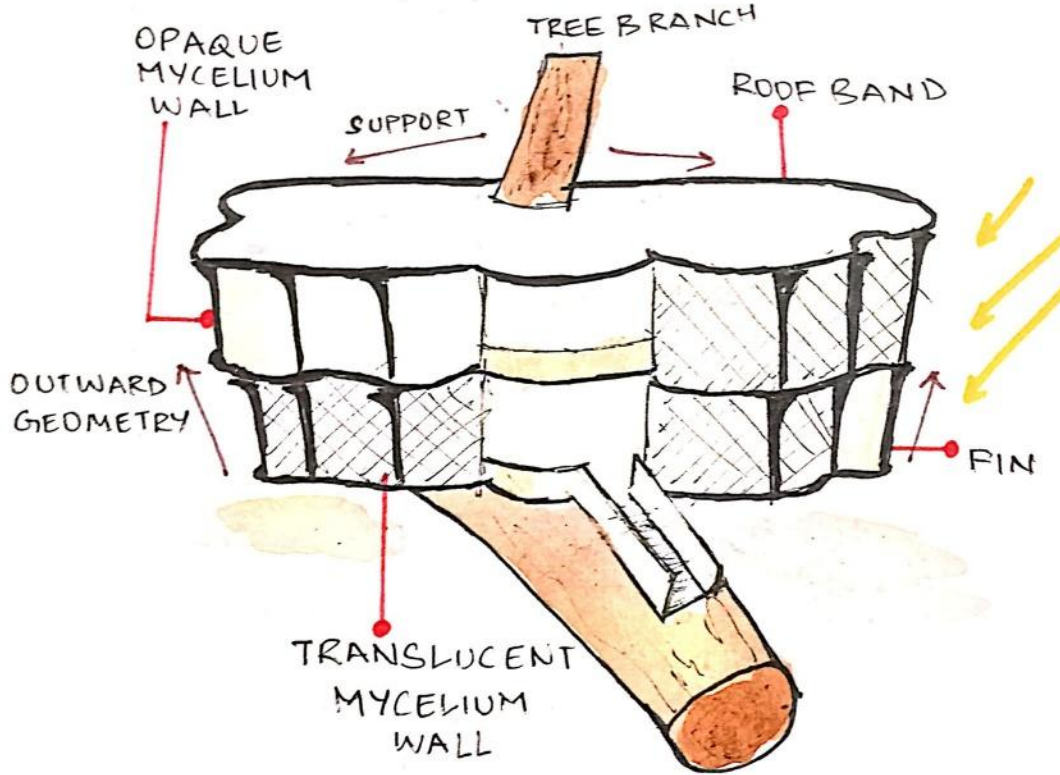
- 1.The oversized flora and fauna make the forest floor unsafe for human habitation.
- 2.The habitat requires camouflage to reduce the risk of detection by giant wildlife.

Site Conditions

- 1.The dense canopy allows only dispersed daylight to reach the site.
- 2.The surrounding vegetation significantly reduces wind speeds.
- 3.Rainwater is intercepted by the canopy and released gradually, often as concentrated drips from large leaves.

Design Response

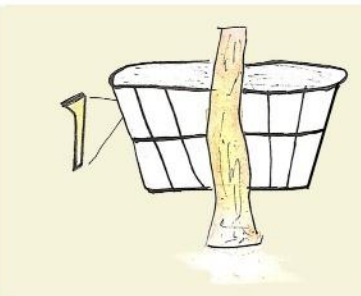
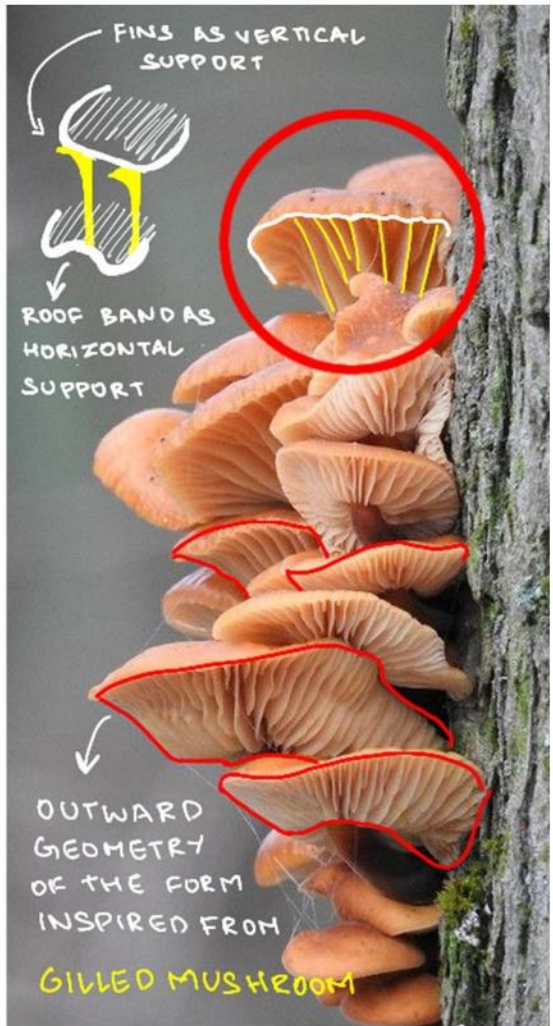
1. **Elevated habitat:** The residence is located within the tree canopy, providing safety from ground-level wildlife also creating a community within a single tree.
2. **Biomimetic form:** Inspired by mushrooms, the building blends into the forest canopy, enhancing camouflage and allowing it to coexist with its surroundings.
3. **Material strategy:** Translucent mycelium facade panels maximize daylight, while opaque panels provide privacy where required.
4. **Roof response:** The outward-sloping upper level functions as a protective canopy, directing rainwater away from the lower floors. It sheds runoff, reducing direct water exposure.



Why Mushrooms?

Nature offers countless forms of shelter—bird nests, burrows, termite mounds, and tree hollows. However, each of these announces the presence of a home, making it easier for predators to locate vulnerable occupants. A mushroom offers a different strategy. It is a quiet participant in the forest, naturally integrated with its surroundings through its form, texture, and colour. It neither dominates the landscape nor advertises the presence of life within. On Planet B, where every organism is three times larger than on Earth, blending into the ecosystem becomes a critical survival strategy. Inspired by this, the house is designed not to conquer the forest, but to coexist with it, living as a quiet participant rather than competing for attention.

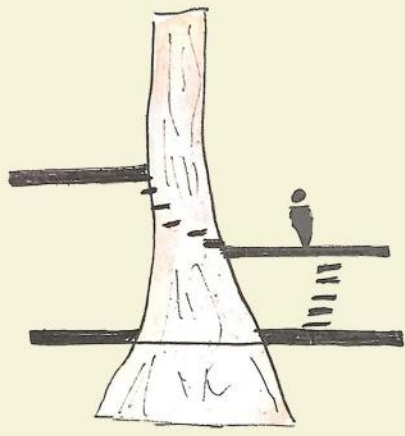
FORM SKETCHES



The overall form is derived from the gilled mushroom, whose outward-opening geometry creates the building's conical profile. The gills of the mushroom inspire the rib-like structural framework, forming the primary structural system of the facade. The mushroom's outward-sloping cap inspires the continuous structural band that wraps around the building, directing loads while reinforcing the overall form.

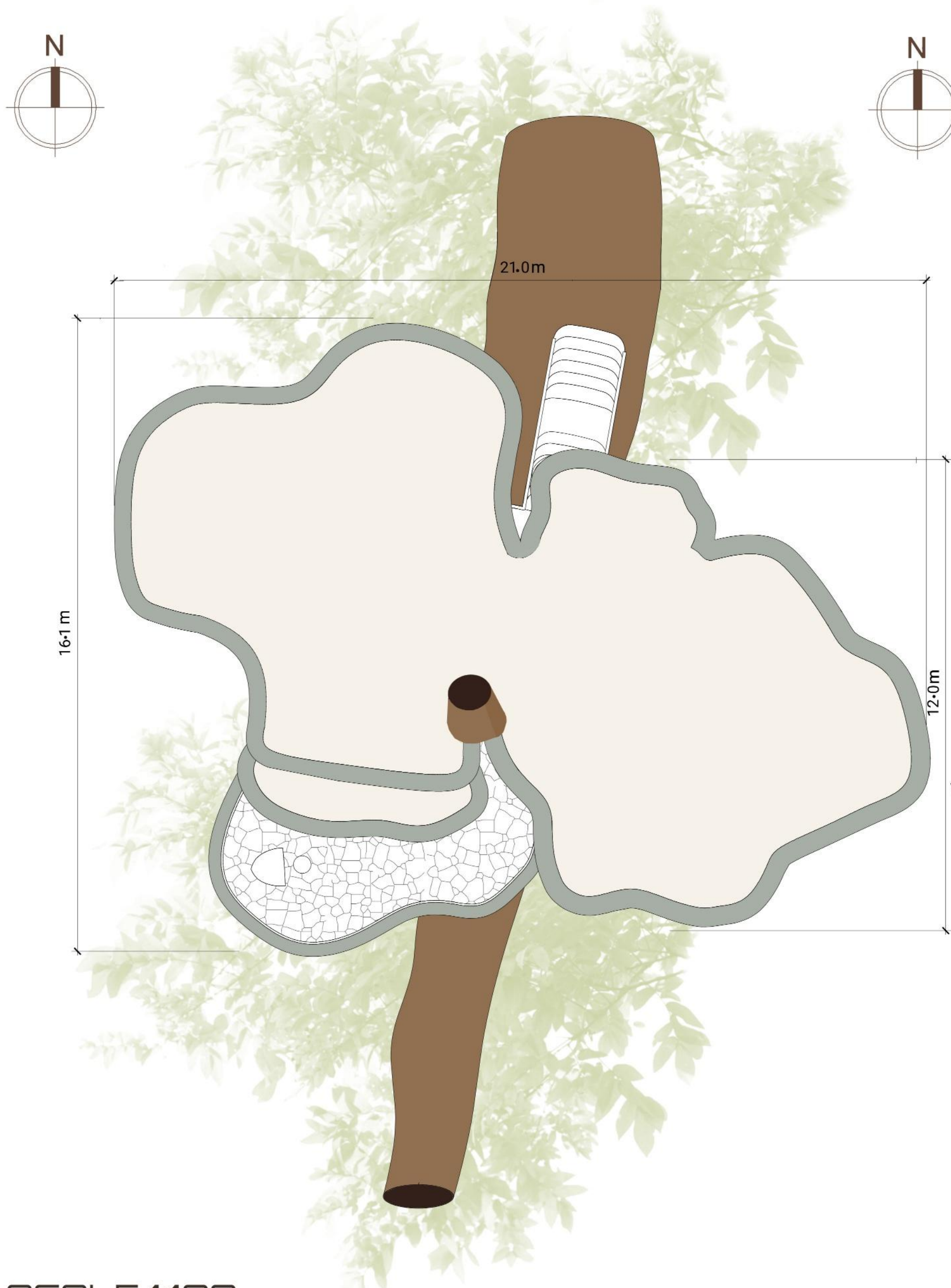


The split-level organisation reflects the layered growth of fungi (Polypore Aka Bracket fungi), creating a continuous spatial journey where movement, views, and nature unfold gradually rather than all at once.



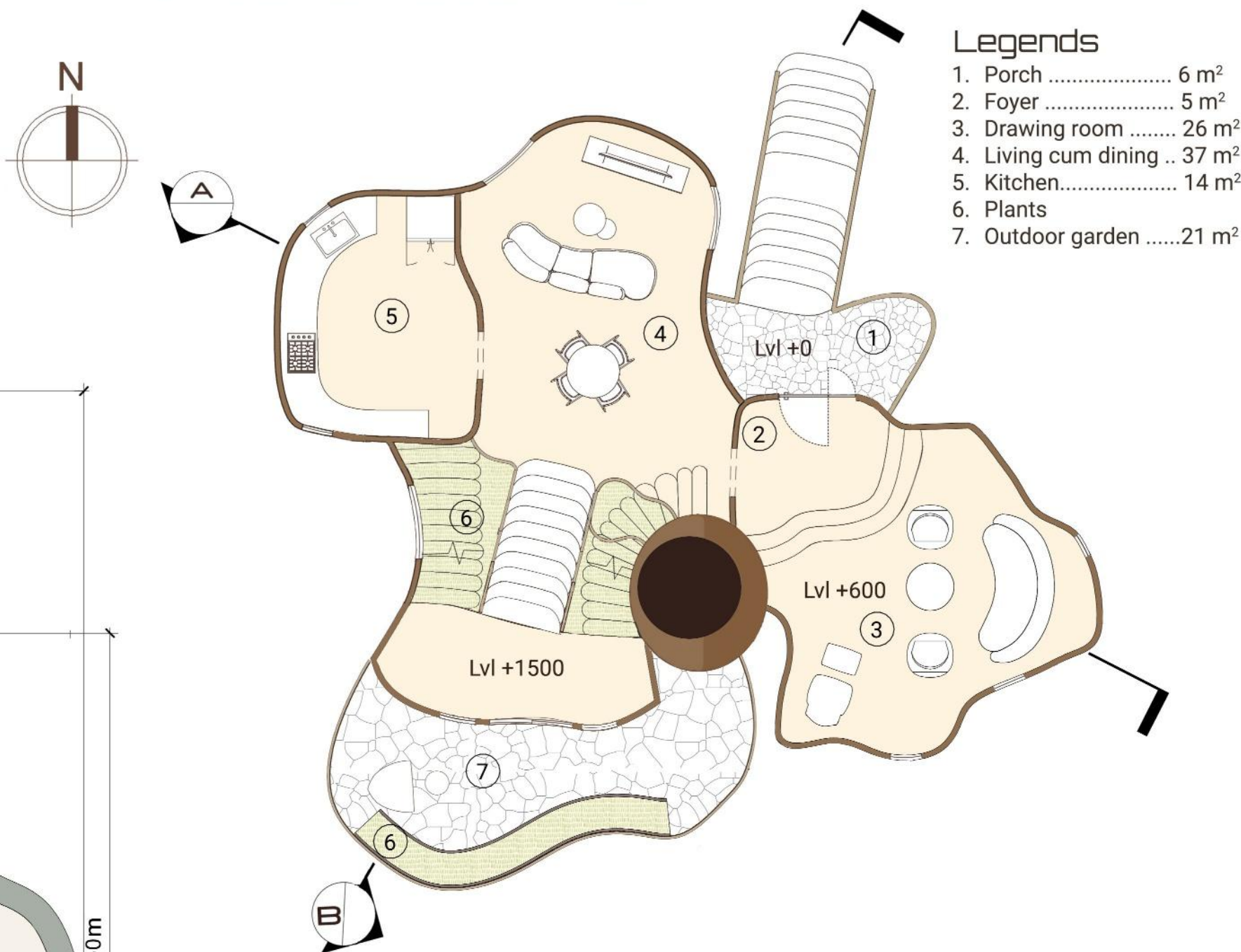
MYCELIUM HOUSE

SITE PLAN

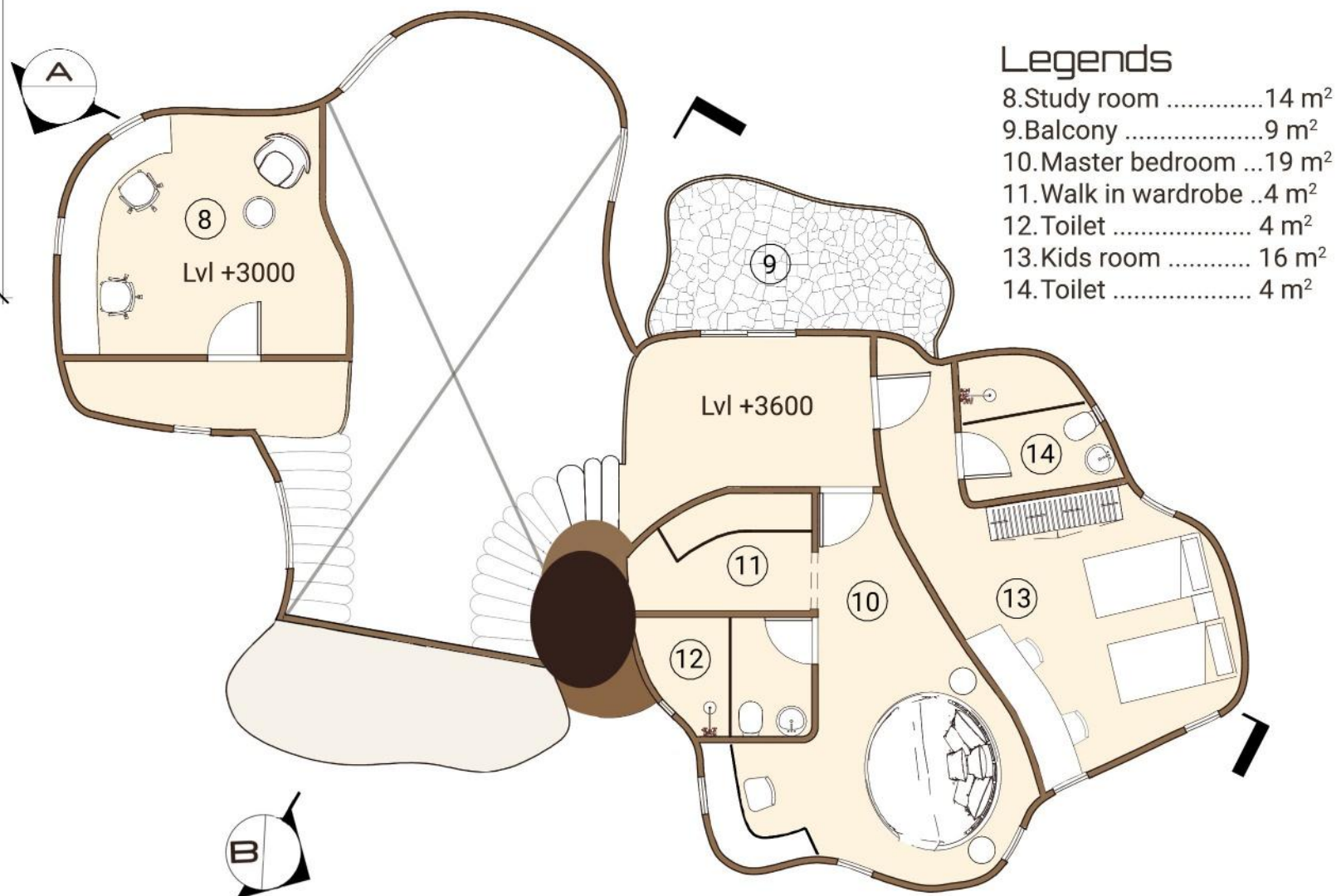


SCALE 1:100

GROUND FLOOR PLAN

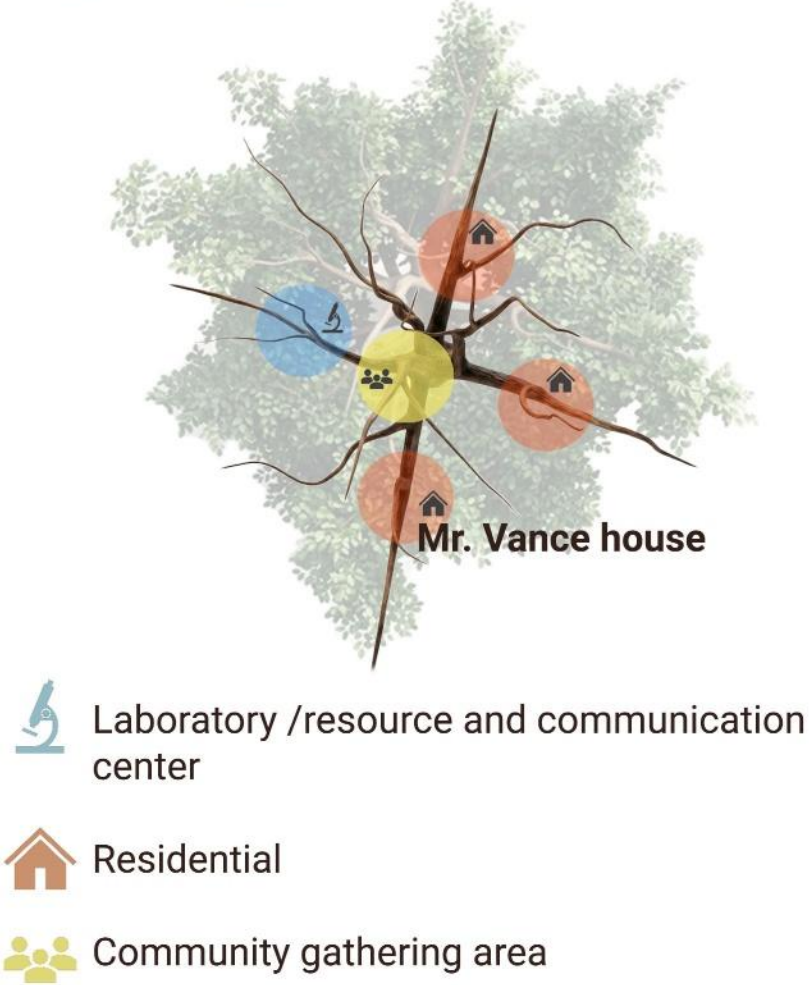


FIRST FLOOR PLAN

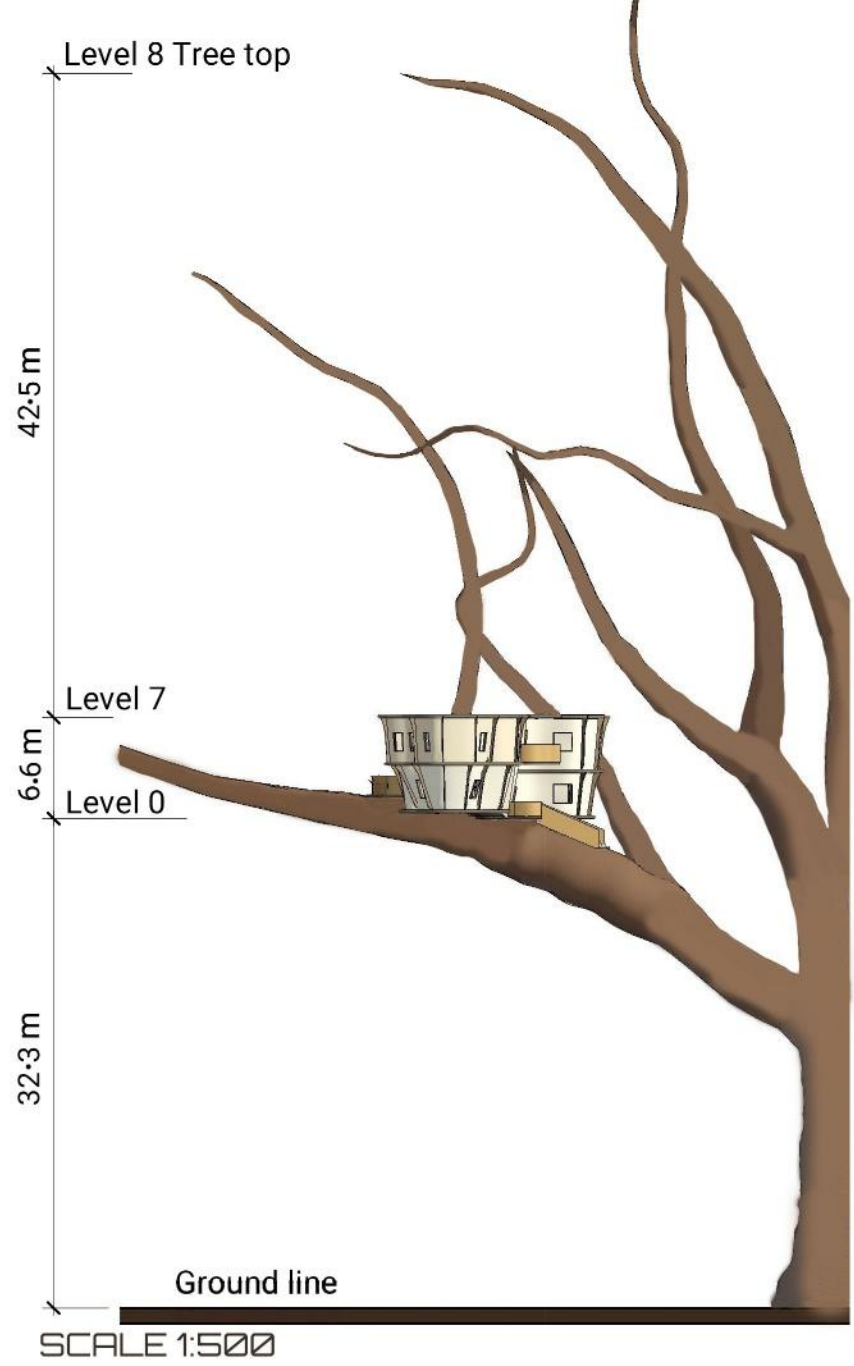


FLOOR PLANS

KEY PLAN



EAST ELEVATION

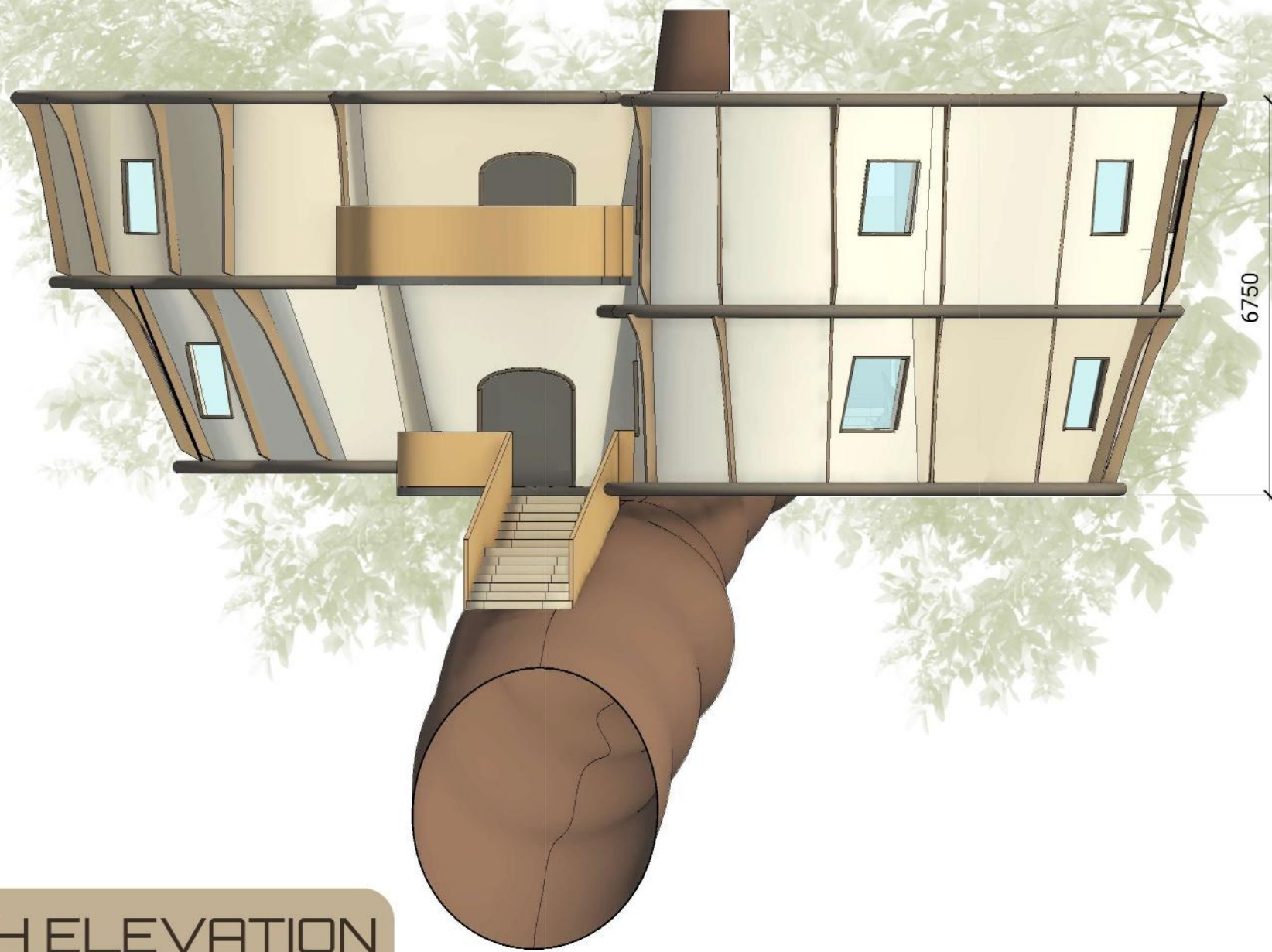


MYCELIUM HOUSE

SECTION A A'



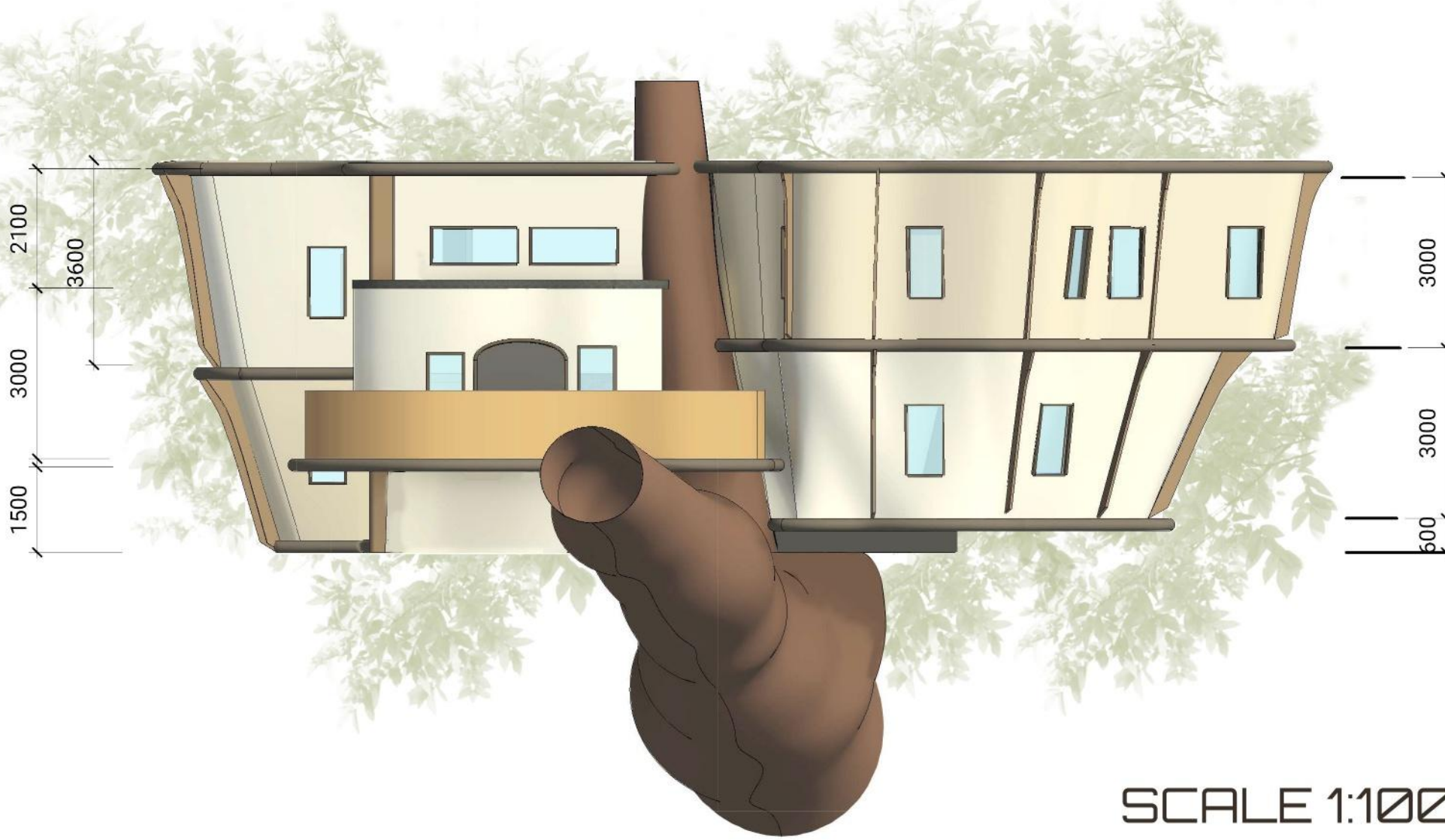
NORTH ELEVATION



SECTION B B'



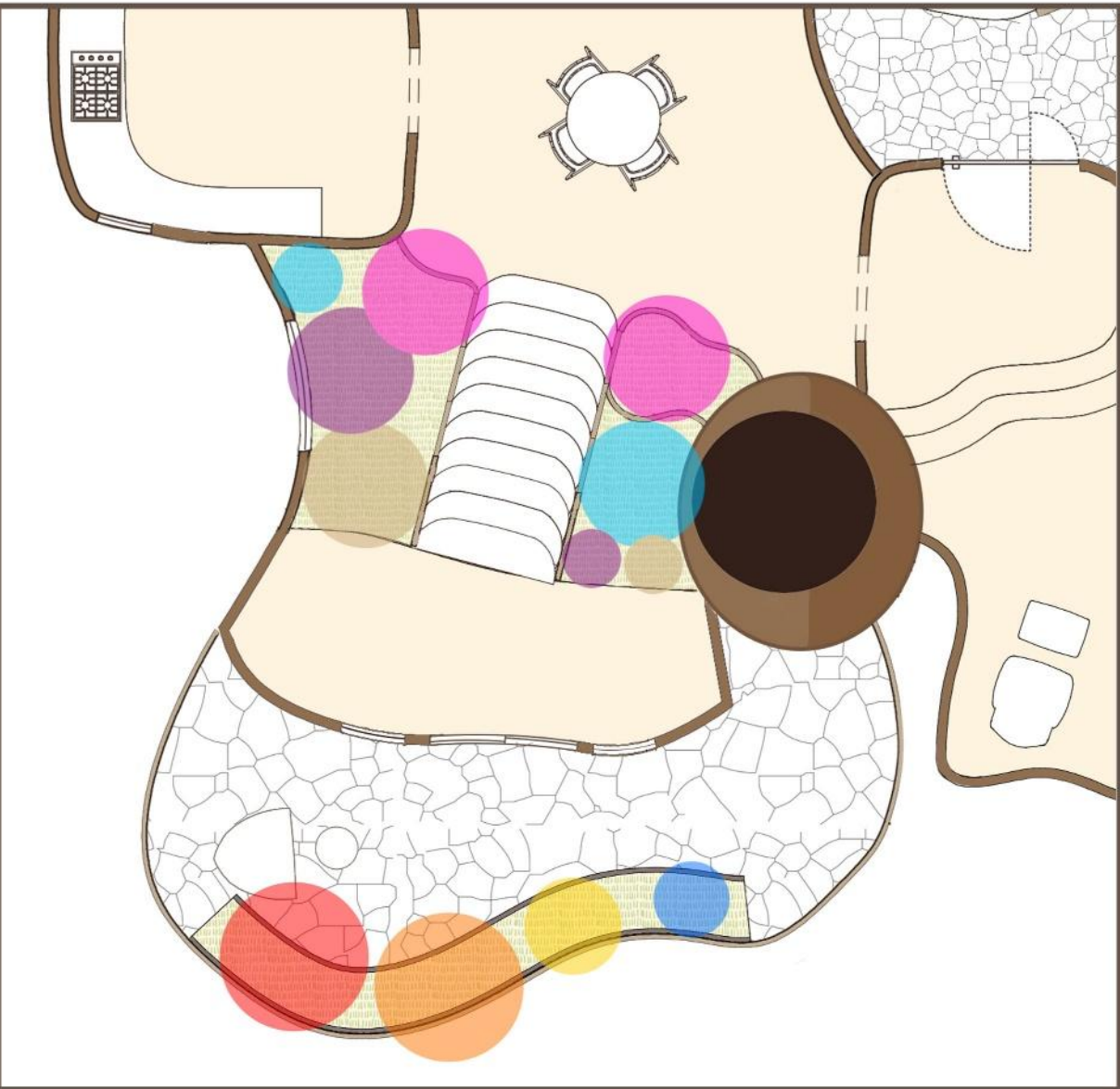
SOUTH ELEVATION



SCALE 1:100

MYCELIUM HOUSE

LANDSCAPE PLAN



- Crops – Carrot
- Leafy Greens – Lettuce
- Climbing Crops – Pole Beans
- Fruiting Crops – Tomato
- Feature Foliage - Monstera
- Forest Ferns - Bird's Nest Fern
- Trailing Vines - Pothos
- Air-Purifying Plants - Snake Plant

The landscape is designed as an extension of the family's mission on Planet B. While the outdoor Earth Crop Trial Garden serves as a living laboratory for cultivating and studying edible Earth crops, the indoor Garden creates a calming atmosphere that reconnects the occupants with familiar nature, transforming the house into both a place of scientific discovery and a comforting home.

Outdoor Earth Crop Trial Garden

The Earth Crop Trial Garden serves as a living research laboratory where Mrs. Vance, a microbiologist, cultivates and studies edible Earth crops to evaluate their adaptability, growth, and productivity within Planet B's ecosystem. Organised into dedicated crop categories, the garden supports the family's scientific mission while promoting self-sufficiency through fresh food production. Beyond its functional role, the edible garden enriches the landscape by introducing seasonal colours, textures, and changing plant forms. It transforms the outdoor space into a productive and interactive environment, seamlessly integrating scientific exploration with everyday family life.



Evaluates root development and soil compatibility on Planet B.



Tests fast-growing edible crops for continuous food production.



Maximizes food production using vertical supports while minimizing ground area.



For studying flowering, fruit production, and long-term adaptability.

Indoor Garden

The Indoor Biophilic Garden is designed to create a calming and restorative living environment through a carefully curated selection of foliage plants that thrive in filtered natural light. The planting palette strengthens the visual connection between the interiors and the surrounding forest while enhancing indoor environmental quality and occupant well-being.



Thrives in bright, filtered light and its large foliage increases humidity while creating a lush indoor environment.



Well suited to humid, low-light interiors, making it ideal for reinforcing the forest-like atmosphere of the home.

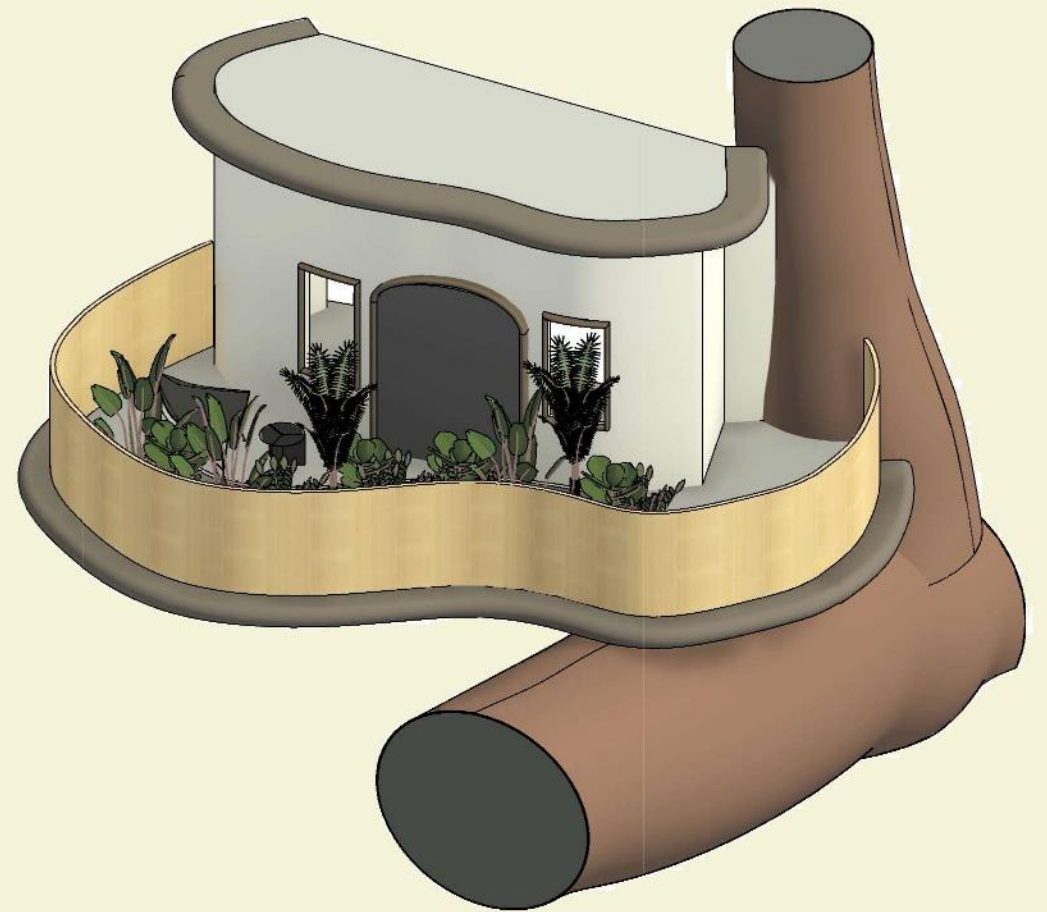


Highly adaptable to indoor conditions, it helps improve indoor air quality while thriving in indirect light.

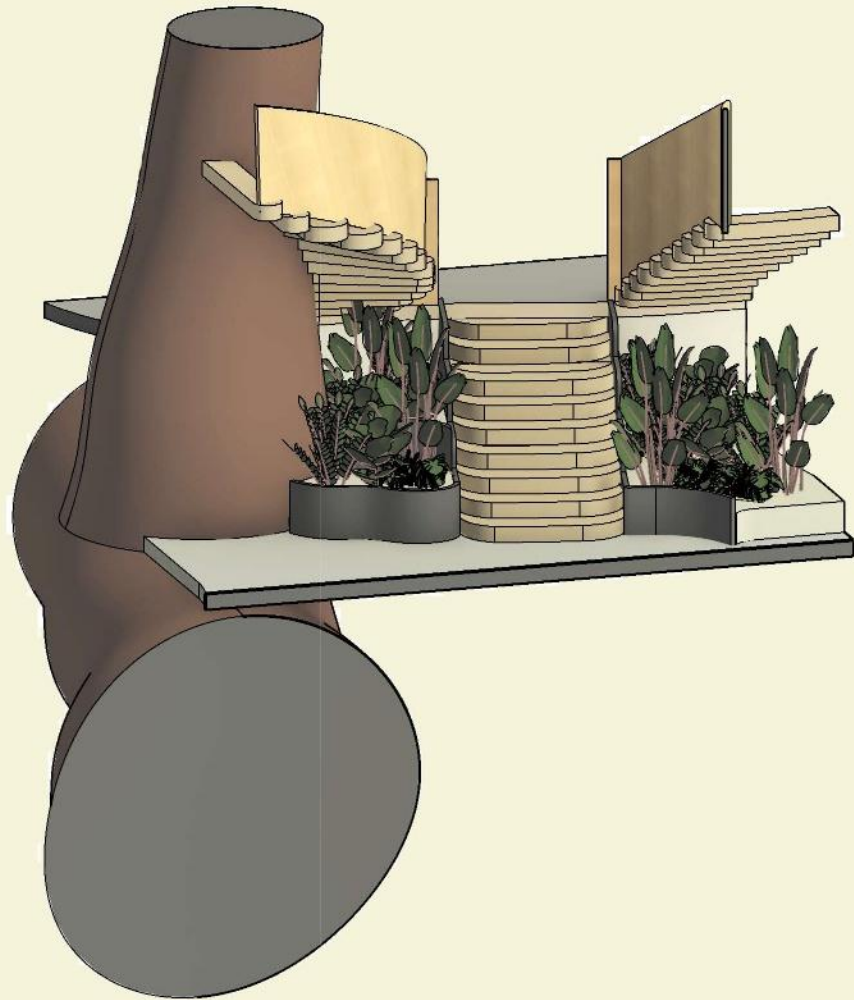


Known for its air-purifying qualities and exceptional tolerance to low light, making it ideal for low-maintenance indoor planting.

VIEW OF OUTDOOR GARDEN



VIEW OF INDOOR GARDEN



VIEWS



MYCELIUM HOUSE

The form channel rainwater away from the facade, protecting the mycelium envelope while directing runoff towards the surrounding landscape.

The translucent mycelium envelope diffuses natural daylight deep into the interior, providing soft, evenly distributed illumination throughout

Mycelium-composite block
Used for Roof/ring band structural and densest

Mycelium-treated wood
Used for Fins-Structural

Opaque Compressed mycelium panel
Used for Bedroom/kitchen walls

Glass
Used for Window openings

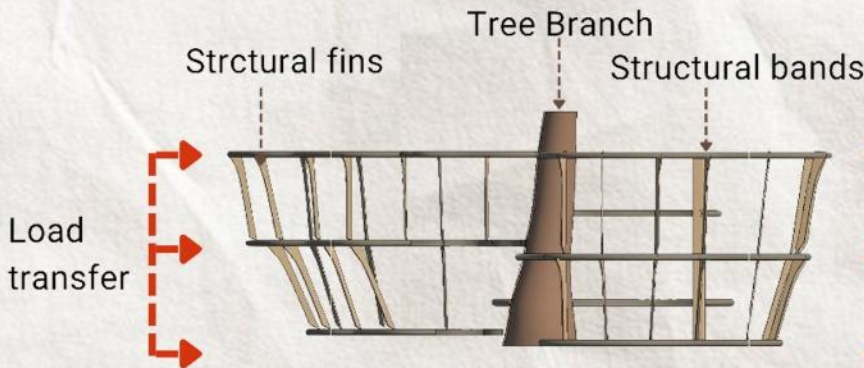
Translucent Raw mycelium membrane
Used for Social space walls - lets light through

The continuous mycelium envelope integrates opaque and translucent panels, providing insulation, filtered daylight, and a unified biomaterial expression throughout the house.

Openings and the central double-height volume promote natural ventilation, thermal comfort and allows natural light to come in.

The house is designed as a quiet extension of its environment, where every architectural element works together to support life on Planet B. Rather than relying on mechanical systems, the design uses natural light, ventilation, and water management to create comfortable living spaces. The use of mycelium as the primary building material further strengthens the relationship between architecture and nature, resulting in a home that is efficient, resilient, and seamlessly integrated into its forest setting.

STRUCTURAL SYSTEM



Vertical loads travel from the roof down through the tapered fins. It is thick near the band, thin at the outer rim, so material matches actual load. Each floor's fins hand their load off to a continuous ring band running around that level, which acts like a compression hoop, spreading the combined weight evenly rather than concentrating it at any single point. The band then meets the trunk, distributing the entire floor's load across a broad area of bark. This is what carries both the straight-down weight and the tipping force of the cantilever making it the structural system. Wind load is less due to the forest canopy protecting the trees.

MATERIAL

A single base material, grown mycelium. It appears at different densities depending on structural and functional demand: raw and translucent where light matters most, compressed and opaque for privacy, and densest where load is highest

CONCLUSION

The proposal reimagines the relationship between architecture and nature on Planet B, where survival depends on coexistence rather than control. Inspired by the ecological behaviour of mushrooms, the design combines biomaterial innovation, passive environmental strategies, and family-centred spaces to create a home that is resilient, sustainable, and deeply connected to its surroundings. More than a shelter, it represents a future where humanity learns to live with nature rather than against it.



EXPLOADED VIEW